

INTERNATIONAL CONFERENCE ON RECENT TRENDS IN ADVANCED COMPUTING
2019, ICRTAC 2019

Implementation Topology of Full Adder Cells

Rekib Uddin Ahmed, Prabir Saha *

National Institute of Technology Meghalaya, Shillong, Meghalaya 793003, India

Abstract

Design of different complementary metal-oxide-semiconductor (CMOS) topologies of 1-bit full adder (FA) cell in terms of power, delay, power-delay-product (PDP) and energy-delay-product (EDP) have been analyzed for the application of the intelligent systems. The FA cells have been implemented with respect to the different CMOS topologies such as complementary-CMOS (C-CMOS), complementary pass-transistor logic (CPL), CMOS transmission gates (TGA), and CMOS transmission function (TFA) and simulated at 180 nm technology. The work has been extended to the comparison of FA cell implementation using the increasingly demanding ultra-thin-body silicon-on-insulator (UTBSOI) transistors. Based on the simulation results, the application of UTBSOI in the FA cell has resulted in 26.7%, 24.3%, 44.6% and 44.7% improvements over that of best design architecture reported in this literature in terms of power, delay, PDP, and EDP, respectively.

© 2019 The Authors. Published by Elsevier B.V.

This is an open access article under the CC BY-NC-ND license (<http://creativecommons.org/licenses/by-nc-nd/4.0/>)

Peer-review under responsibility of the scientific committee of the INTERNATIONAL CONFERENCE ON RECENT TRENDS IN ADVANCED COMPUTING 2019.

Keywords: BSIM-IMG; C-CMOS; delay; full adders; power; TFA; TGA; UTBSOI.

1. Introduction

Nowadays, the trend of increasing mobility and performance in portable battery-operated devices has resulted in the downscaling of technology node in which small on-chip area, high speed, and low power consumption are the primary concern [1]. In modern systems, the microprocessors and digital signal processors rely on efficient arithmetic operations like convolution, correlation, filtering, etc. Full adders (FA) are the prime necessity to do these operations. The power consumption in any systems can be reduced by reducing the energy consumed per arithmetic operation. The circuits involved in an arithmetic operation can be made to consume less power by lowering the

* Corresponding author. Tel.: 9436594902

E-mail address: sahaprabir1@gmail.com

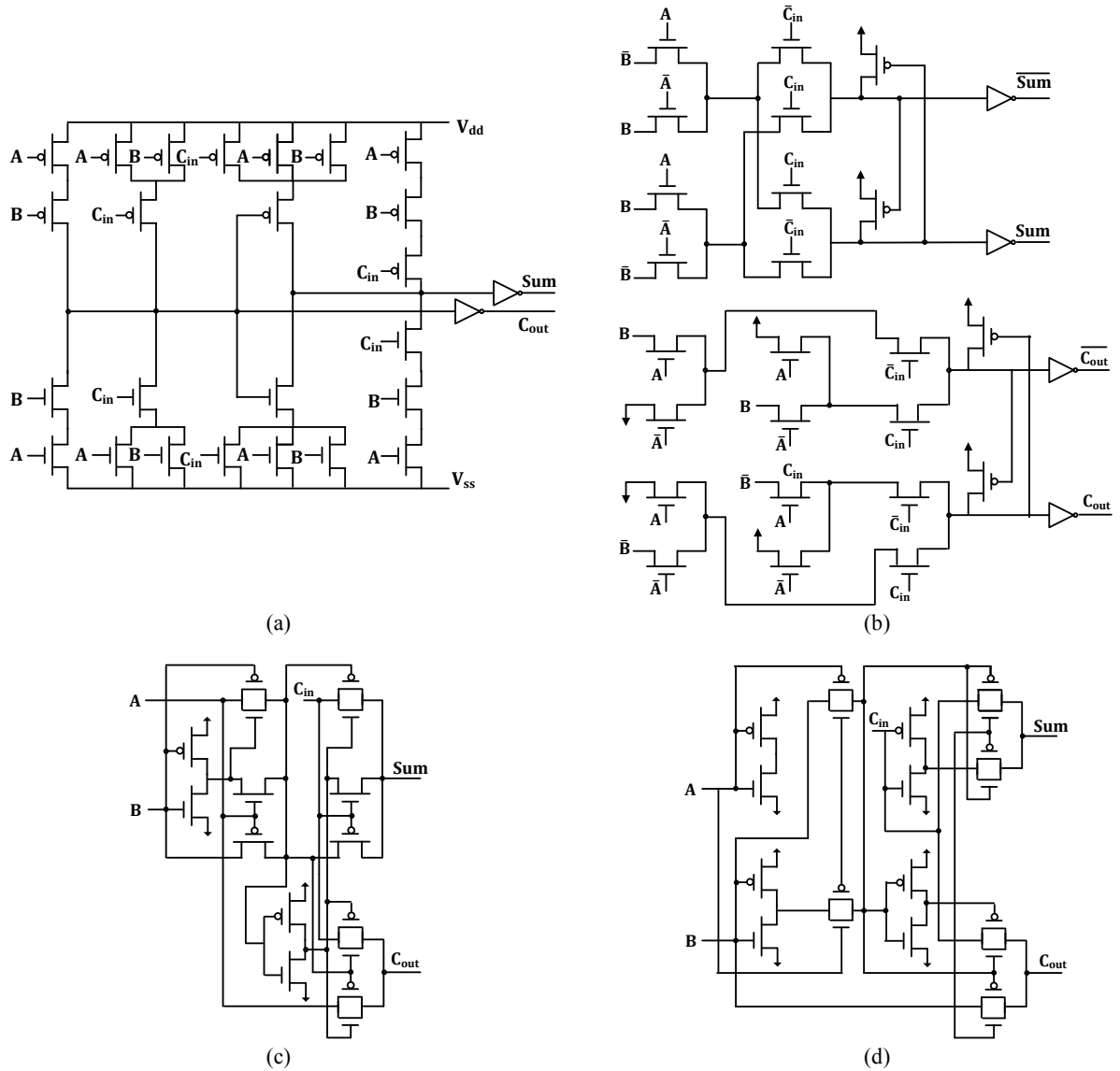


Fig. 1 Different FA topologies : (a) C-CMOS, (b) CPL, (c) TGA, and (d) TFA.

supply voltage (V_{dd}) along-with input pulses at very low frequency (f) which in turn increases the delay and degrade the driving capability of the system [2]. The designs of CMOS FA cells can be classified into two categories [3]: 1) static logic and 2) dynamic logic. Static FA cells are simpler to design and more reliable with lesser power consumption, but the on-chip area requirement is larger as compared to the dynamic FA cells. Different logic topologies have been used to design the FA cells. Each topology has its advantages and disadvantages, tend to favour one performance aspect at the expense of others. Therefore, the design of the FA cells with high speed (less delay) and low power consumption has become a primary issue in the design of such systems. The standard topologies for the 1-bit FA cell are as follows:

- The static complementary CMOS (C-CMOS) FA [4] as shown in Fig. 1(a) is based on the conventional CMOS design in which the Boolean expressions are realized by p-channel (pMOS) pull-up and n-channel (nMOS) pull-down MOSFETs, and it requires 28 transistors.

- The complementary pass-transistor logic (CPL) FA [3] as shown in Fig. 1(b), requires 32 transistors to implement the Boolean expression of FA. The disadvantage of CPL topology is the threshold voltage (V_{th}) drop while passing the logic level at inputs to the output and inverters are required to ensure the drivability at the output.
- The CMOS transmission gates FA (TGA) [5,6] as shown in Fig. 1(c), requires 20 transistors and, is based on transmission gates (TG).
- The CMOS transmission function FA (TFA) [7,8] as shown in Fig. 1(d), requires 16 transistors and, is based on the transmission function theory.

The different topologies of FA cell implement the same Boolean expressions, but their way of generating the intermediate nodes and producing the outputs are different; thus, the transistor count varies significantly.

This paper presents a comparison of performance parameters like power, delay, PDP, and EDP achieved by the different FA topologies considered namely C-CMOS, CPL, TGA, and TFA. The transistor-level designs of the FA cells are simulated in Cadence-spectre using the generic process-design kit (gpdn) 180 nm technology. Among the FA cells (C-CMOS, CPL, TGA, and TFA), the TFA has been found as the most promising topology by exhibiting the least PDP of 0.0290 fJ at $V_{dd} = 1.2$ V and frequency of input pulses $f = 20$ MHz with rising and falling time (t_{rf}) = 10 ps. The comparison of performance parameters is also extended to TFA designed using the ultra-thin-body silicon-on-insulator (UTBSOI) transistor. The use of UTBSOI in TFA has resulted in 26.7%, 24.3%, 44.6% and 44.7% improvements over that of CMOS based in terms of power, delay, PDP, and EDP, respectively.

Table 1: Transistor's parameters in case1 and case 2.

Parameters	Case 1		Case 2	
	nMOS	pMOS	nMOS	pMOS
Channel length (L)	180 nm	180 nm	180 nm	180 nm
Channel width (W)	2 μ m	2 μ m	540 nm	1.08 μ m
Source diffusion area	1.2×10^{-12} m ²	1.2×10^{-12} m ²	486×10^{-15} m ²	972×10^{-15} m ²
Drain diffusion area	1.2×10^{-12} m ²	1.2×10^{-12} m ²	486×10^{-15} m ²	972×10^{-15} m ²
Source diffusion periphery	5.2 μ m	5.2 μ m	2.88 μ m	3.96 μ m
Drain diffusion periphery	5.2 μ m	5.2 μ m	2.88 μ m	3.96 μ m
Threshold voltage (V_{th})	0.548 V	− 0.516 V	0.519 V	− 0.504

2. Comparison of FA cells

The simulation of the FA cells based on C-CMOS, CPL, TGA, and TFA topologies are performed in Cadence-spectre with the gpdn 180 nm technology. The simulations are carried out considering two cases: 1) case 1, where the generic values of transistor's parameter set are considered, 2) case 2, where the optimized values are used as given in Table 1. At the initial stage, the simulations are performed at different V_{dd} ranging from 0.8 V to 1.8 V [2,6,13] with the frequency of input pulses (f) is set at 20 MHz [3] with $t_{rf} = 10$ ps. Different values of V_{dd} are chosen to compare the variations in performance parameters (delay, power, PDP, and EDP) of the topologies considered with respect to V_{dd} . Table 2 shows the performance parameters achieved by the C-CMOS, CPL, TGA, and TFA FA cells in case 1. The variations in performance parameters with respect to V_{dd} are shown in Fig. 2. From Fig. 2(a), it can be observed that the power consumption in the FA cells increases with increase in V_{dd} . Whereas, the delay seems to decrease with increasing V_{dd} . As it has been stated earlier that the power consumption of any system can be reduced by reducing the V_{dd} , but it also increases the delay [2,13] hence gets proved from Fig. 2(a) and (b). In case 1, the least PDP = 0.0653 fJ is exhibited by the TFA (TFA 1) at $V_{dd} = 1.2$ V. The power consumption in CPL is higher than the other topologies due to the high switching activity of intermediate nodes, presence of static inverters, and overloading of its inputs. The performance parameters achieved in case 2 are given in Table 3. By optimizing the transistor's parameter values in case 2, it is expected to reduce the power consumption of all FA cells so as to achieve a minimum PDP. The performance parameter metrics are improved in case 2 as it can be observed from Fig. 3. The power consumption, delay, and PDP exhibited by the FA cells in case 2 are reduced by an average of 49.57%, 13.5%, and 53.97%, respectively than that of FA cells in case 1.

Table 2: Performance parameters achieved by the FA cells in case 1 at different supply voltages.

Power consumption (μW):				
Power supply (V)	0.8	1.0	1.2	1.8
C-CMOS	0.2788	0.4561	0.6778	1.675
CPL	0.6601	1.082	1.612	3.995
TGA	0.3417	0.5451	0.7939	1.819
TFA	0.2318	0.3652	0.5244	1.166
Delay (ns)				
C-CMOS	1.42058	0.66397	0.42019	0.2101
CPL	0.91018	0.4391	0.28528	0.1537
TGA	0.4876	0.23526	0.15071	0.07246
TFA	0.38162	0.18839	0.12454	0.0938
PDP (fJ)				
C-CMOS	0.3960	0.3028	0.2848	0.3519
CPL	0.6008	0.4751	0.4598	0.6140
TGA	0.1666	0.1282	0.1196	0.1318
TFA	0.0884	0.0688	0.0653	0.1093
EDP (fJ $\times$ ns)				
C-CMOS	158.3946	121.108	113.9135	140.767
CPL	240.2875	190.0864	183.9485	245.6126
TGA	66.6549	51.2866	47.8654	52.7146
TFA	35.3761	27.5237	26.1160	43.7389

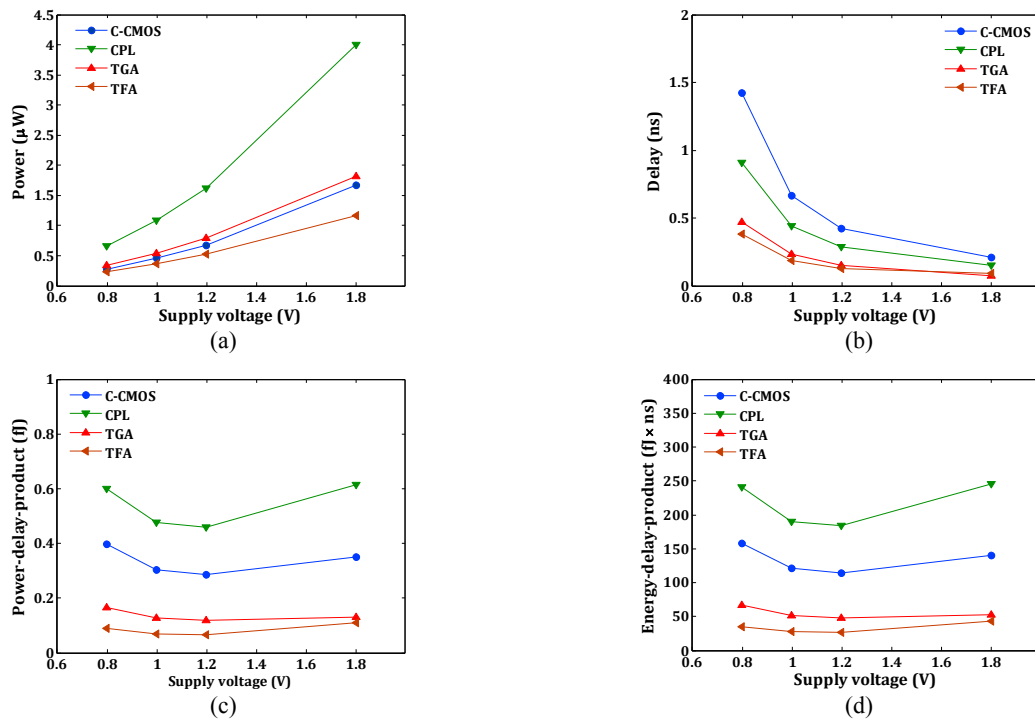


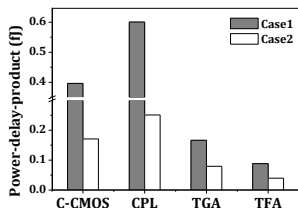
Fig. 2 The variation of (a) power, (b) delay, (c) PDP, and (d) EDP of the FA cells in case 1 with respect to supply voltage.

The TFA in case 2 (TFA 2) is exhibiting reduced PDP for all supply voltages. The comparison of PDP and EDP achieved by the FA cells in case 2 is shown in Fig. 4, from where it is observed that the reduced PDP and EDP achieved by TFA 2 approaches the least value at $V_{dd} = 1.2$ V. The simulation results also reveal that among the topologies considered, the TFA 2 exhibits the best performance and the $V_{dd} = 1.2$ V can be considered as the optimum supply voltage for carrying out the rest part of the simulation. Now, considering the $V_{dd} = 1.2$ V, the TFA 2 is simulated at different values of f [1,3,6,13] to observe its change in performance parameters with respect to

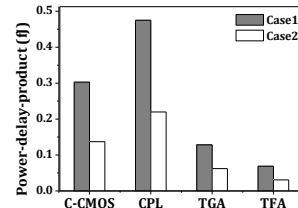
Table 3: Performance parameters achieved by the FA cells in case 2 at different supply voltages.

Power consumption (μW):

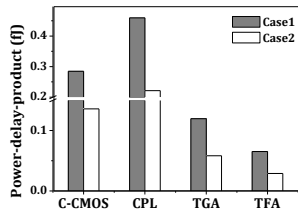
Power supply (V)	0.8	1.0	1.2	1.8
C-CMOS	0.1428	0.2311	0.3414	0.8334
CPL	0.3114	0.5112	0.7647	1.92
TGA	0.1789	0.2829	0.4106	0.9328
TFA	0.1207	0.1891	0.2715	0.6044
Delay (ns)				
C-CMOS	1.1976	0.59292	0.39794	0.22088
CPL	0.8064	0.43027	0.28948	0.16632
TGA	0.44495	0.21957	0.14192	0.06993
TFA	0.32809	0.16223	0.10691	0.05685
PDP (fJ)				
C-CMOS	0.1710	0.1370	0.1358	0.1840
CPL	0.2511	0.2199	0.2213	0.3193
TGA	0.0796	0.0621	0.0583	0.0652
TFA	0.0396	0.0307	0.0290	0.0344
EDP (fJ $\times$ ns)				
C-CMOS	68.4188	54.8154	54.3586	73.6413
CPL	100.4774	87.9902	88.5519	127.7171
TGA	31.7382	24.8333	23.2890	26.1468
TFA	15.8434	12.2743	11.6104	13.7463



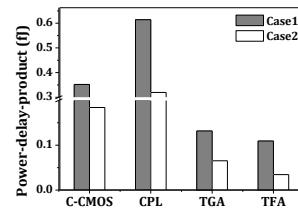
(a)



(b)

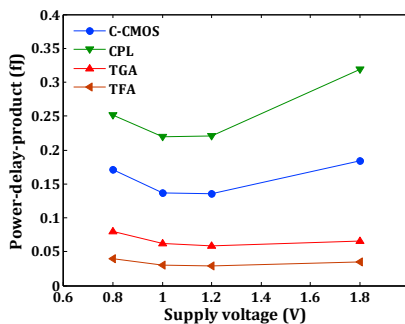


(c)

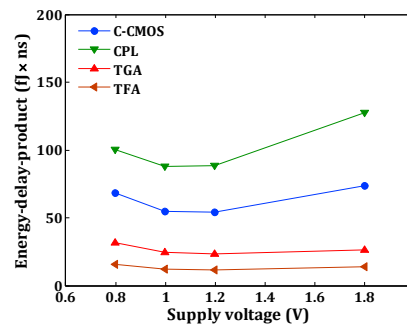


(d)

Fig. 3. Comparison of PDP of the FA cells in case 2 at different supply voltages: (a) 0.8V, (b) 1.0 V, (c) 1.2 V, (d) 1.8 V.



(a)



(b)

Fig. 4. Variation of (a) PDP, (b) EDP of the FA cells in case 2 with respect to supply voltages.

Table 4: Performance parameters achieved by the TFA cell in case 2 (TFA 2) at different frequencies.

Frequency (MHz)	20 ($t_{rf} = 10$ ps)	50 ($t_{rf} = 4$ ps)	80 ($t_{rf} = 2.5$ ps)	100 ($t_{rf} = 2$ ps)
Power consumption (μ W):	0.2715	0.6922	1.114	1.395
Delay (ns):	0.10691	0.1039	0.1031	0.1029
PDP (fJ):	0.0290	0.0719	0.1148	0.1435
EDP (fJ $\times$ ns):	11.6104	11.5121	11.4853	11.4836

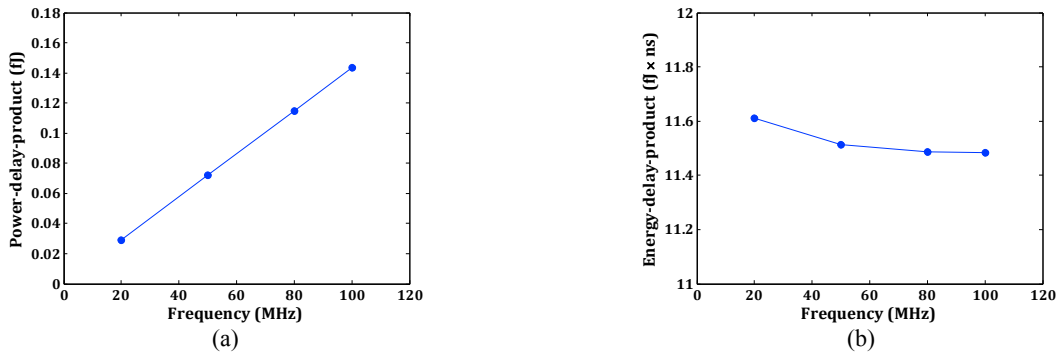


Fig. 5. Variation of (a) PDP, (b) EDP of the TFA cells in case 2 (TFA 2) at different frequencies.

frequency. Table 4 shows the performance parameters achieved by TFA 2 at different frequencies. The PDP achieved by TFA 2 increases with increasing the f as it can be seen from Fig. 5(a), whereas the EDP remains fairly constant with increasing f as can be seen from Fig. 5(b). To exploit the benefits of UTBSOI technology, the TFA 2 is replicated using the UTBSOI transistors.

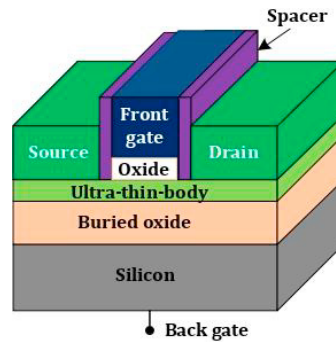


Fig. 6. Schematics showing 3-dimensional view of a UTBSOI transistor.

3. The UTBSOI Transistor

The process variation parameters of conventional MOSFETs have become a major cause in degrading the performance of CMOS based integrated circuits (ICs). The short-channel effects (SCE) are becoming prone as the device dimension is scaled down to the nanometer regime. Due to the SCEs in MOSFET, a large abnormality is observed in the ICs produced by the present manufacturing process. To overcome the limitations of SCEs, multi-gate MOSFETs like double-gate MOSFET, FinFET, and UTBSOI have been proposed [9]. The UTBSOI transistor, as shown in Fig. 6 is a dual gated transistor has better scalability and superior controllability of gates over the shorter channel region. In some low-power and high-performance applications, MOSFETs with different threshold voltages are required in the same IC. The UTBSOI offers the flexibility of multiple threshold voltage control by the use of its back-gate bias [10] which was earlier achieved by using different channel doping concentrations in conventional MOSFETs [11]. To utilize the benefits of UTBSOI, an accurate and fast computation model for

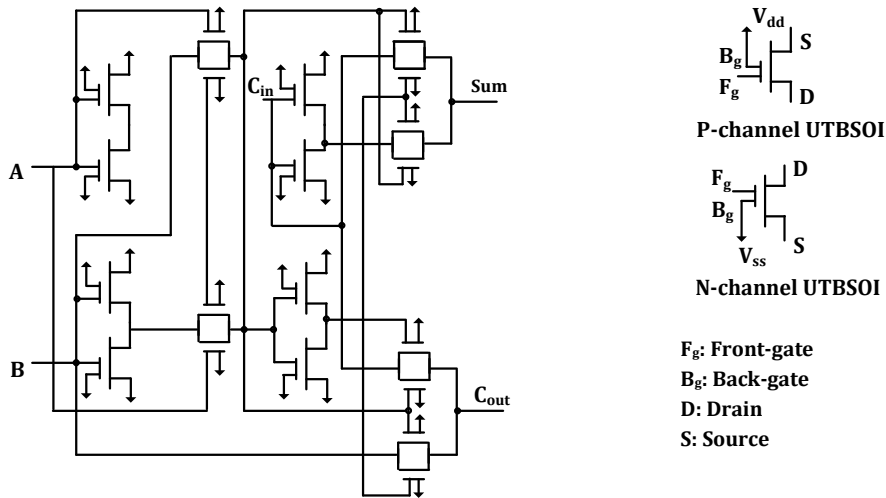


Fig. 7. Schematics showing the UTBSOI based TFA 2.

Table 5. Summary of performance parameters achieved by the CMOS and UTBSOI based TFA 2 considering $V_{dd} = 1.2$ V at $f = 20$ MHz.

Designs of TFA 2	CMOS	UTBSOI	% Change
Power consumption (μ W)	0.2715	0.1984	26.7
Delay (ns)	0.10691	0.08087	24.3
PDP (fJ)	0.0290	0.01604	44.6
EDP(fJ $\times$ ns)	11.6104	6.4186	44.7

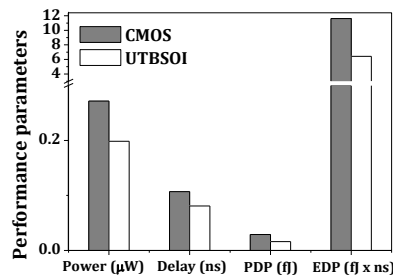


Fig. 8. Comparison of performance parameters achieved by the CMOS and UTBSOI based TFA 2.

the device is required. BSIM-IMG [12] is the industry-standard model for the UTBSOI which have been validated with the hardware silicon-based data from multiple technologies.

The UTBSOI based TFA 2 cell is simulated by connecting the back gates of p- and n-channel UTBSOI transistors to V_{dd} and V_{ss} , respectively [Fig. 7]. In order to compare the performance parameters of the CMOS and UTBSOI based TFA 2 cell, the simulations are carried out considering $V_{dd} = 1.2$ V, $f = 20$ MHz. Table 5 shows the performance parameters achieved by TFA 2 in UTBSOI technology. The TFA 2 based on UTBSOI technology has 26.7%, 24.3%, 44.6%, and 44.7% improvement in terms of power, delay, PDP, and EDP, respectively over that CMOS based. The improvement in performance parameters due to the use of UTBSOI can also be observed from the graphical comparison given in Fig. 8. The area comparison is not provided in the analysis since the layout of UTBSOI based TFA 2 is not possible due to the non-availability of the process-design kit (pdk) for the UTBSOI in Cadence-spectre [14].

4. Conclusion

In this paper, a comparison of the different topologies of the CMOS FA cell is presented. The topologies considered for comparison are C-CMOS, CPL, TGA, and TFA. The simulation of the topologies is carried out in Cadence-spectre using the gpdn 180 nm technology. The simulation process is performed considering two cases, where in the first case (case 1), the generic values of transistor's parameters are considered and in the second case (case 2), the optimized value of the transistor's parameters are used. Among the FA topologies considered in case 1, TFA cell (TFA 1) has the minimum PDP. Performance parameters achieved by the FA topologies are improved in case 2, and from the comparison of performance parameters from case 1 and 2, it is found that the least PDP is exhibited by the TFA in case 2 (TFA 2). In other words, TFA 2 has the best performance over the other topologies considered. The design of TFA 2 is replicated using the UTBSOI transistors in order to compare the performances achieved in CMOS and UTBSOI technology. The UTBSOI based TFA 2 design exhibits 44.6% less PDP than that of CMOS based TFA 2. From the simulation results and comparison of performance parameters, it can be concluded that the UTBSOI transistors can successfully be used to design such FA cells. Investing considerable effort in developing a pdk for the UTBSOI would be a welcome step in enabling further improvements in digital systems.

Acknowledgements

This work is supported by the Ministry of Electronics and Information Technology (MeitY), Government of India, under Visvesvaraya PhD Scheme.

References

- [1] P. Bhattacharyya, B. Kundu, S. Ghosh, V. Kumar, and A. Dandapat. (2015) "Performance analysis of a low-power high-speed hybrid 1-bit full adder circuit." *IEEE Transactions on VLSI Systems* **23**(10): 2001–2008.
- [2] K. Navi et al. (2009) "A novel low-power full-adder cell for low voltage." *Integration. VLSI Journal* **42**(4): 457–467.
- [3] R. Zimmermann and W. Fichter. (1997) "Low-power logic styles: CMOS versus pass-transistor logic." *IEEE Journal of Solid-State Circuits* **32**(2): 1079–1090.
- [4] A.M. Shams, T.K. Darwish, and M.A. Bayoumi. (2002) "Performance analysis of low-power 1-bit CMOS full adder cells." *IEEE Transactions on VLSI Systems* **10**(1): 20–29.
- [5] N. Weste and K. Eshraghian. (1993) *Principles of CMOS VLSI Design, A Systems Perspective*, 2nd ed. Reading, MA: Addison Wesley.
- [6] M. Alioto and G. Palumbo. (2002) "Analysis and comparison on full adder block in submicron technology." *IEEE Transactions on VLSI Systems* **10**(6): 806–823.
- [7] N. Zhuang and H. Wu. (1992) "A new design of the CMOS full adder," *IEEE Journal of Solid-State Circuits* **27**(5): 840–844.
- [8] X. Wu and F. Prosser. (1988) "Design of ternary CMOS circuits based on transmission function theory," *International Journal of Electronics* **65**(5): 891–905.
- [9] N. Paydavosi et al. (2013) "BSIM-SPICE models enable FinFET and UTB IC designs." *IEEE Access* **1**: 201–215.
- [10] Q. Liu et al. (2010) "Ultra-thin-body and BOX (UTBB) fully depleted (FD) device integration for 22 nm node and beyond." In *Symposium on VLSI Technology*, Honolulu, Hawaii, 15–17 Jun. 2010, 61–62.
- [11] S. Khandelwal et al. (2012) "BSIM-IMG: A compact model for ultrathin-body SOI MOSFETs with back-gate control." *IEEE Transactions on Electron Devices* **59** (8): 2019–2026.
- [12] BSIM-IMG. [Online]. (Accessed on 2017, Aug 10). Available: <http://bsim.berkeley.edu/models/bsimimg>
- [13] P. Kumar and R.J. Sharma. (2016) "Low voltage high performance hybrid full adder." *Engineering Science and Technology, an International Journal* **19**(1): 559–565.
- [14] C. Hu et al. (2019) *Industry Standard FDSOI Compact Model BSIM-IMG for IC Design*, 1st ed. Woodhead Publishing.